BEYOND MINI-BATCH RELATIONS: HEAT-DIFFUSION DISTILLATION OF TEACHER EMBEDDING MANIFOLDS

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ABSTRACT

Compact text embedding models power semantic search, clustering, and retrieval-augmented generation, and distillation from a strong teacher is the standard way to train them. The dominant relational objective matches teacher and student similarities inside a mini-batch, but a mini-batch is a uniform sample of the corpus: an anchor's semantic neighbors rarely appear in it, so the local geometry of the teacher space is supervised at a vanishing rate. We propose **HeatGeo**, which replaces batch relations with anchored local supervision. HeatGeo builds a mutual k-nearest-neighbor graph over cached teacher embeddings, treats it as a discrete semantic manifold, and trains the student to match a family of heat-diffusion distributions at scale-dependent resolutions over sampled candidate neighborhoods, using only final teacher embeddings—no hidden states, online teacher passes, or task labels. We support this design with theory: mini-batch relational objectives provably undersample local neighborhoods; matching one-step diffusion locally controls multi-step global geometry; diffusion-weighted sampling approximates full neighborhoods at a cost set by target sharpness rather than neighborhood size; and the diffusion loss leaves exactly one similarity offset per anchor undetermined, which supervising the rows a teacher random walk visits—nodes the mini-batch has already encoded—collapses to one offset per connected component of the teacher graph. Across nine datasets and three teacher–student pairs, HeatGeo attains the best overall average among compact students in two of three settings and second best in the third, with the largest gains on semantic textual similarity.

1 INTRODUCTION

Text embedding models have become a basic infrastructure layer for semantic search, clustering, reranking, and retrieval-augmented generation. Their usefulness comes not only from individual vectors, but from the geometric structure those vectors induce: neighborhoods, semantic regions, and smooth paths between related examples. Strong recent embedding models, including LLM-based encoders, often learn rich semantic geometry but are costly to run at scale. Encoding large private collections, refreshing indexes as documents change, or serving retrieval on constrained hardware therefore motivates compact student encoders that inherit the geometry of a stronger teacher.

Knowledge distillation offers a practical route to this compression (Hinton et al., 2015). In language representation learning, distillation has been applied through output matching, hidden-state alignment, self-attention transfer, and compact pretrained architectures (Sanh et al., 2019; Jiao et al., 2020; Wang et al., 2020; Xu et al., 2024). For embedding models, common objectives include pointwise alignment between student and teacher embeddings, contrastive learning with teacher-derived positives, teacher-anchored layer alignment, and relational losses that match pairwise similarities within a mini-batch. These objectives provide valuable supervision at different granularities: individual examples, internal layers, sampled pairs, or local batch structure.

The geometric view suggests a complementary form of supervision. A teacher embedding space can be treated as a sampled semantic manifold: nearby examples define local neighborhoods and multi-hop paths reveal semantic continuity. Pointwise losses encourage coordinate agreement, layer-alignment methods improve internal transfer under capacity mismatch, and mini-batch relational losses preserve sampled similarity patterns. HeatGeo instead asks whether the student can learn the teacher's manifold geometry directly from diffusion targets derived from the teacher embedding space.

This motivates the central question of this work: can a compact student inherit the teacher's semantic manifold by matching heat-style diffusion on a teacher-induced neighborhood graph? If the teacher embeddings define a discrete manifold over training examples, then random walks on this manifold reveal how semantic mass flows from each anchor to nearby examples, broader semantic regions, and higher-order communities. Matching this diffusion process gives the student geometry-aware supervision that is not tied to teacher internals or a particular student architecture.

We propose **HeatGeo**, a heat-diffusion manifold distillation method for compact text embedding models. HeatGeo first caches teacher embeddings and constructs a mutual k-nearest-neighbor graph using teacher cosine similarity. This graph acts as a discrete approximation to the teacher's semantic manifold. From it, HeatGeo defines a transition matrix and precomputes diffusion targets across a range of diffusion times, from the ambient teacher metric itself to one-step, two-step, and longer-range semantic random walks. During training, each anchor example is paired with a candidate set containing high-diffusion neighbors, teacher hard negatives, and random negatives, and candidate sets are pooled across the mini-batch so that anchors are compared on a common set of columns. The student then learns neighborhood distributions over those columns and minimizes a weighted KL divergence to the teacher targets. A second term extends supervision beyond the anchors: random walks on the teacher graph select non-anchor nodes that the batch has already encoded, and the student is trained to reproduce the teacher's one-step kernel at those nodes as well, which both densifies supervision and pins down the similarity offsets that anchor-only matching leaves free.

A single design decision makes this family more than target smoothing: the student matches each scale at its own resolution, which keeps the scales distinct constraints on the student instead of collapsing them into one smoothed target (Proposition 3). HeatGeo needs no pointwise regression, contrastive pair construction, layer alignment, or graph-coordinate objective, and adds no trainable parameters beyond the student encoder.

Our contributions are:

- We formulate text embedding distillation as manifold geometry transfer, using the teacher embedding space as a discrete semantic manifold.
- We develop a theory that traces the reasoning chain from the failure mode of batch-relational supervision to each component of the objective: why supervision must be anchored to local neighborhoods, why matching them locally is enough to control global geometry, why small sampled candidate sets suffice, and why a single calibration term completes the loss (Section 3).
- We introduce HeatGeo, which distills a diffusion-time family of heat-diffusion distributions from a teacher-induced mutual kNN graph into a compact student encoder, requiring neither teacher hidden states nor task labels, with resolution-dependent student temperatures that keep the scales informative.

2 Related Work

Text Embedding Models and Evaluation. Text embedding models encode variable-length text into fixed-dimensional vectors for retrieval, clustering, semantic textual similarity, and transfer learning. Sentence-BERT made transformer encoders practical for sentence-level similarity search by using siamese and triplet-network training (Reimers & Gurevych, 2019), while SimCSE showed that contrastive learning with dropout noise or entailment supervision can produce strong sentence embeddings (Gao et al., 2021b). Evaluation suites such as BEIR and MTEB further shifted the field from single-task similarity scores toward broad retrieval, classification, clustering, and semantic textual similarity evaluation (Thakur et al., 2021; Muennighoff et al., 2023). Continued maintenance of MTEB reflects the need for reproducible, multi-task embedding evaluation as models and datasets evolve (Chung et al., 2025).

Recent embedding systems increasingly rely on larger foundation models, richer training mixtures, and specialized retrieval objectives. BGE-M3 combines multilingual, multi-functionality, and multi-granularity training with self-knowledge distillation across dense, sparse, and multi-vector retrieval modes (Chen et al., 2024). Qwen3 Embedding scales embedding and reranking models from Qwen3 foundation models (Zhang et al., 2025b), and Dewey targets long-context embedding for retrieval-augmented generation (Zhang et al., 2025a). This line of work provides stronger teachers and broader evaluation targets, but it does not by itself specify how a compact student should inherit the intrinsic geometry of the teacher embedding space. HeatGeo treats these models as black-box embedding teachers: it requires only their final embeddings, then distills diffusion geometry induced by their neighborhoods.

Distillation for Language and Embedding Models. Knowledge distillation transfers behavior from a large teacher to a smaller student (Hinton et al., 2015). In pretrained language models, DistilBERT, TinyBERT, and MiniLM compress transformer encoders through combinations of logits, hidden states, attention distributions, and self-attention relations (Sanh et al., 2019; Jiao et al., 2020; Wang et al., 2020). Recent LLM distillation extends this toolkit with on-policy training, contrastive objectives, sparse-logit acceleration, and score-matching formulations (Agarwal et al., 2024; Ko et al., 2025; Anshumann et al., 2025; Kim et al., 2025; Xu et al., 2024). Cross-tokenizer methods such as DSKD and CDM address mismatches between teacher and student vocabularies or tokenization schemes (Zhang et al., 2024b; Chen et al., 2025).

Embedding-specific distillation adapts these ideas to retrieval-oriented representations. Gecko distills LLM-generated and LLM-relabelled retrieval data into compact text embedding models (Lee et al., 2024). Jasper and Stella use multi-stage embedding distillation and Matryoshka Representation Learning to improve both quality and dimensional flexibility (Zhang et al., 2024a). LEAF aligns compact embedding students to teacher representations and supports asymmetric retrieval in which queries and documents may be encoded by different models (Vujanic & Rueckstiess, 2025). Jina Embeddings v5 combines task-targeted distillation with contrastive training to produce compact multilingual embeddings robust to truncation and quantization (Akram et al., 2026). These methods establish distillation as a central recipe for embedding model development. HeatGeo is complementary to this line: rather than introducing new teacher internals or task labels, it converts cached final embeddings into manifold diffusion targets.

Local and Relational Supervision. Pointwise embedding distillation directly matches a projected student embedding to the corresponding teacher embedding. This is simple and architecture-agnostic, but it asks a compact student to reproduce the teacher's absolute coordinates even when the student's capacity and embedding dimension differ. Contrastive sentence-embedding distillation methods such as DistilCSE improve the training signal by transferring contrastive behavior and pairwise preferences (Gao et al., 2021a; Xu et al., 2023). Relational knowledge distillation further argues that students should preserve relations among examples rather than only individual outputs (Park et al., 2019). In embedding training, this idea commonly becomes mini-batch similarity matching: teacher and student pairwise similarity matrices are compared over the examples that happen to co-occur in a stochastic batch.

These local objectives better reflect retrieval than pure coordinate regression, but their supervision is bounded by the sampled batch or pair set. They do not explicitly model multi-hop semantic paths, community structure, or the way probability mass should spread from an anchor across the teacher manifold. Token-level and layer-level alignment methods, including CKA-style representation matching, optimal-transport alignment, EMO-style objectives, and teacher-anchored layer alignment, provide useful intermediate supervision when teacher internals or student layers are available (Kornblith et al., 2019; Alqahtani et al., 2021; Dao et al., 2026). In particular, teacher-anchored layer alignment targets internal transfer and capacity mismatch, while HeatGeo uses only the neighborhood geometry induced by final teacher embeddings.

Graph, Manifold, and Diffusion Perspectives. Graph-based manifold learning studies how neighborhood graphs reveal structure in high-dimensional data. Diffusion Maps use Markov diffusion on a graph to expose multi-scale connectivity and geometry (Coifman & Lafon, 2006). Knowledge distillation on graphs also studies how structural information can be transferred between graph models or from graph teachers to simpler students (Tian et al., 2025; Zhang et al., 2022). These works show that graph diffusion can encode information unavailable from isolated examples.

HeatGeo draws on this graph-manifold view but applies it to a different distillation setting. The graph is not an observed input graph; it is induced by a strong text embedding teacher, and its diffusion distributions become soft manifold-neighborhood targets for the student. This gives compact text encoders a single explicit geometry-aware supervision signal derived from the teacher embedding manifold.

3 From Mini-Batch Relations to Manifold Diffusion

This section develops the reasoning chain behind HeatGeo. First, relational objectives computed inside a mini-batch undersample the teacher's local geometry, and their in-batch targets are biased estimators of neighborhood structure (Section 3.2). Anchored supervision on each example's own neighborhood avoids both defects, and matching local one-step diffusion is enough: it controls the teacher's multi-step, global geometry (Section 3.3). Local neighborhoods are large, but they need not be enumerated—diffusion-weighted sampling approximates them at a cost governed by the sharpness of the diffusion targets rather than by the neighborhood size (Section 3.4). Finally, distribution matching leaves exactly one similarity offset per anchor undetermined; supervising the rows at non-anchor nodes, which a random walk selects and the batch has already encoded, collapses that freedom to one offset per connected component of the teacher graph (Section 3.5). All statements are proved in Appendix A; Section 5.3 instantiates the bounds and tests their predictions.

3.1 Setup

Let $\mathcal{D} = \{x_i\}_{i=1}^N$ be the distillation corpus with cached teacher embeddings t_i on the unit sphere, and write $c^T_{ij} = \cos(t_i, t_j)$. Let $\mathcal{N}_k(i)$ denote the k nearest neighbors of x_i under c^T. A neighborhood graph on $\mathcal{D}$ induces a row-stochastic transition matrix P, with lazy walk $\tilde{P} = \frac{1}{2}(I+P)$, and the teacher's diffusion family consists of the rows $p_{i,r} = \delta_i^\top \tilde{P}^r$ for $r \geq 1$ together with an ambient distribution $p_{i,0} \propto \exp(c^T_{ij}/\tau_0)$. The student produces embeddings s_i with scores $c^S_{ij} = \cos(s_i, s_j)$ and, on a column set Ω, distributions $q_{i,r} \propto \exp(c^S_{ij}/\tau_r)$. Training minimizes the restricted, weighted divergence

$$\mathcal{L}_{\mathrm{diff}} = \frac{1}{N} \sum_i \sum_r \omega_r \, D_{\mathrm{KL}}\Big(p_{i,r}\big|_{C_i} \,\Big\|\, q_{i,r}\Big), \tag{1}$$

where C_i is a per-anchor candidate set and $p|_C$ denotes restriction to C followed by renormalization. Section 4 specifies the graph, the candidate sets, and the temperatures; the analysis here depends only on the abstract structure.

For distributions p, q on the same finite set, $\mathrm{TV}(p, q) = \frac{1}{2}\|p - q\|_1$ denotes total variation.

3.2 Mini-Batch Relations Undersample Local Geometry

Relational distillation compares teacher and student similarities over the examples that co-occur in a stochastic mini-batch (Park et al., 2019). A mini-batch, however, is a uniform sample of the corpus, and local geometry lives on $\mathcal{N}_k(i)$, a set of measure k/N. The first result quantifies how rarely the two meet. We model an epoch as a uniformly random partition of $\mathcal{D}$ into batches of size B, resampled independently across epochs.

Proposition 1 (Neighborhood coverage)**.** *Fix an anchor i and let $\mathcal{B} \ni i$ be its batch. Then (i)* $\mathbb{E}\,|\mathcal{B} \cap \mathcal{N}_k(i)| = (B-1)\,k/(N-1)$*; (ii)* $\Pr[\mathcal{B} \cap \mathcal{N}_k(i) = \emptyset] \geq 1 - (B-1)\,k/(N-1)$*; (iii) across T independently resampled epochs, the expected fraction of pairs $\{(i,j) : j \in \mathcal{N}_k(i)\}$ that ever co-occur is at most* $T(B-1)/(N-1)$.

When $B \ll N/k$, most parameter updates contain no neighbor of the anchor at all, and supervision of the anchor's neighborhood accrues at rate $O(B/N)$ per epoch—a property of the sampling scheme, independent of the loss or the optimizer—so observing each neighborhood even once requires on the order of N/B epochs.

Coverage alone could in principle be repaired by training longer. The next result shows that for in-batch *normalized* targets the defect is a bias, which no amount of training repairs. It requires only that the teacher's ambient distribution is concentrated on the neighborhood it is meant to describe.

Assumption 1 (Target concentration)**.** There is $\eta \in [0,1)$ such that $p_{i,0}(\mathcal{N}_k(i)) \geq 1-\eta$ for every anchor i.

Assumption 1 holds whenever neighbor cosines exceed non-neighbor cosines by a margin Δ and $\tau_0 \lesssim \Delta/\log(N/k\eta)$ (Appendix A.2).

Proposition 2 (Bias of in-batch targets)**.** *Let $p_i^{T,\mathcal{B}}(j) \propto \exp(c_{ij}^T/\tau_0)$ for $j \in \mathcal{B}\setminus\{i\}$ be the in-batch teacher target. Under Assumption 1,*

$$\mathbb{E}_{\mathcal{B}}\Big[\mathrm{TV}\Big(p_i^{T,\mathcal{B}},\ p_{i,0}\Big)\Big] \ \geq\ (1-\eta)\left(1-\frac{(B-1)k}{N-1}\right). \tag{2}$$

The bound does not decay with the number of epochs: it is a property of the estimator, not of its variance. On the event that the batch misses the neighborhood, the normalized target places all of its mass on unrelated examples, and the student is taught a spurious neighborhood. Together, Propositions 1 and 2 say that batch-relational supervision estimates the bulk of the similarity distribution well and the local geometry poorly—which motivates anchoring supervision to each example's own neighborhood.

3.3 Matching Local Diffusion Transfers Global Geometry

Anchored supervision raises the converse question: if the student only ever matches *local* one-step distributions, what does it learn about the *global* geometry—multi-hop connectivity, communities, diffusion distances? This subsection shows the local constraints are sufficient: one-step control propagates to all diffusion times with at most linear error growth.

We interpret the graph as a discrete manifold in the standard sense of diffusion maps (Coifman & Lafon, 2006): under suitable sampling conditions, neighborhood-graph diffusion converges to the heat semigroup of the underlying manifold (von Luxburg et al., 2008; Trillos & Slepčev, 2018; Calder & Trillos, 2022). We use this only as interpretation; all statements below are about the discrete objects.

Lemma 1 (One step to r steps)**.** *Let P, Q be row-stochastic matrices with lazy walks $\tilde{P}, \tilde{Q}$, and suppose $\max_i \mathrm{TV}(P_i, Q_i) \leq \varepsilon$. Then for every $r \geq 1$, $\max_i \mathrm{TV}\big(\delta_i^\top \tilde{P}^r,\ \delta_i^\top \tilde{Q}^r\big) \leq r\varepsilon/2$.*

The loss in Eq. 1 controls each row only on a candidate set, so two further ingredients connect the trained objective to the full rows: the candidate set must carry most of the teacher's mass, and the student must not hide mass outside it.

Lemma 2 (Restriction)**.** *Let p, q be distributions on a common finite set and C a subset with $p(C) \geq 1-\delta_T$ and $q(C) \geq 1-\delta_S$. Then $\mathrm{TV}(p,q) \leq \mathrm{TV}\big(p|_C, q|_C\big) + \delta_T + \delta_S$.*

Remark 1 (Why negatives: leakage control). The teacher-side defect δ_T is a property of the candidate construction; the student-side defect δ_S is what negatives target. Split $C = C^+ \cup C^-$ with the teacher's restricted mass on the negative block C^- at most ε_{neg}. Driving the restricted divergence to ε forces the student's restricted mass on C^- below $\varepsilon_{\text{neg}} + \sqrt{\varepsilon/2}$ (data processing and Pinsker; Appendix A.5). Because the negatives are redrawn from the complement of the diffusion support at every epoch, a small restricted loss constrains the student's mass on fresh samples of the complement throughout training—the mechanism that keeps δ_S small. Theorem 1 accordingly takes δ_S as a hypothesis; Appendix A.5 discusses the quantitative extension from sampled columns to the full complement.

Theorem 1 (Local-to-global transfer)**.** *Let P^T and P^S be teacher and student transition matrices on the same support. Suppose for every anchor i: the restricted divergence satisfies $D_{\mathrm{KL}}(p_{i,1}|_{C_i} \,\|\, q_{i,1}) \leq \varepsilon$, the candidate set carries teacher mass $p_{i,1}(C_i) \geq 1-\delta_T$, and the student's full-row mass outside C_i is at most δ_S. Then for every diffusion time $r \geq 1$,*

$$\max_i\ \mathrm{TV}\Big(\delta_i^\top(\tilde{P}^T)^r,\ \delta_i^\top(\tilde{P}^S)^r\Big)\ \leq\ \frac{r}{2}\left(\sqrt{\varepsilon/2}+\delta_T+\delta_S\right). \tag{3}$$

Matching one-step neighborhoods within ε therefore pins down every longer-range diffusion distribution—and with it diffusion distances and community structure—up to an error that grows only linearly in diffusion time. The multi-scale family sharpens this further: supervising a broad scale directly replaces the compounded bound with a per-scale one.

Corollary 1 (Multi-scale tightening)**.** *Under the conditions of Theorem 1 applied at scale r directly (restricted divergence ε_r, mass defects δ_T, δ_S at scale r), the r-step error is at most $\sqrt{\varepsilon_r/2}+\delta_T+\delta_S$, with no factor of r.*

Supervising several scales at once is only informative if the scales remain distinct constraints on the student. The following observation, used to justify resolution-dependent temperatures in Section 4.4, shows that a shared student distribution collapses the family.

Proposition 3 (Collapse under a shared student distribution)**.** *If all scales share one student distribution p_i^S, then*

$$\sum_r \omega_r D_{\mathrm{KL}}\big(p_{i,r}^T \,\|\, p_i^S\big) = H\big(\bar{p}_i^T,\, p_i^S\big) - \sum_r \omega_r H\big(p_{i,r}^T\big), \qquad \bar{p}_i^T = \sum_r \omega_r p_{i,r}^T, \tag{4}$$

where the second term does not depend on the student parameters.

Cross-entropy is linear in its target, so with one shared student distribution any two scale families with the same mixture produce identical gradients, and multi-scale supervision degenerates into target smoothing. Distinct temperatures τ_r break the equivalence and force the student to reproduce the teacher's similarity gaps rather than only its ranking.

Distinct temperatures raise the question of which values they may take, and two members of the family turn out to have no freedom at all: whenever a term's target is itself a softmax of teacher cosines at some temperature, matching it at any other temperature changes what zero loss asks of the student.

Proposition 4 (Temperature ties)**.** *Let Ω be a column set with $|\Omega| \geq 2$, and consider a term with target $\mathrm{softmax}_\Omega(c_{i\cdot}^T/\tau)$ and student distribution $\mathrm{softmax}_\Omega(c_{i\cdot}^S/\tau_S)$. (i) Over score assignments $c_{i\cdot}^S$ the term has infimum 0, attained exactly on the affine family $c_{ij}^S = (\tau_S/\tau)\, c_{ij}^T + a_i$ on Ω — a shift family when $\tau_S = \tau$ and a rescaling otherwise. (ii) Suppose the ambient term at the same row also vanishes, on a set Ω_0 such that $\Omega \cap \Omega_0$ contains columns j, j' with $c_{ij}^T \neq c_{ij'}^T$. Then $\tau_S = \tau$.*

Part (ii) is the operative statement: with the ambient scale in the objective, a softmax-of-cosines target must be matched *at its own temperature*, or the two zero sets are disjoint and the objective acquires an artificial floor. This pins two temperatures that would otherwise look like hyperparameters. The sharpest graph scale's target is the transition row itself — the lazy walk at $r = 1$, off-diagonal renormalized — a softmax of teacher cosines at τ_g, so $\tau_1 = \tau_g$. The walk targets of Section 4.5 are transition rows as well, so $\tau_w = \tau_g$, the tie Proposition 7 assumes. The ambient scale itself uses the single temperature τ_0 on both sides, the same-temperature convention of knowledge distillation (Hinton et al., 2015), which by (i) makes its zero set exactly the shift family of Proposition 6. The only temperatures left free are those of the broader scales.

3.4 Sampled Neighborhoods Suffice

The support of $p_{i,r}$ grows with k and with diffusion time, so scoring every anchor against its full multi-hop neighborhood at every step is not viable. HeatGeo instead scores each anchor against a small candidate set drawn in proportion to diffusion mass, from a support computed by a truncated walk whose error is additive across steps (Lemma 3, Appendix A.9). The remaining question is how many draws the candidate set needs.

Proposition 5 (Sampling error)**.** *Let p be a diffusion target for one anchor, and let the candidate set C be formed by m independent draws from p. Then*

$$\mathbb{E}[\mathrm{TV}(p|_C,\, p)] = \mathbb{E}\big[p(\bar{C})\big] \;\leq\; \min_{t>0}\; \Big(e^{-tm} + \mathrm{Tail}_p(t)\Big), \qquad \mathrm{Tail}_p(t) = \sum_{j:\, p_j < t} p_j. \tag{5}$$

In particular, if p places mass at least c/ρ on each atom of its effective support of perplexity $\rho = e^{H(p)}$, the expected error is at most $e^{-cm/\rho}$.

The sample size that suffices is governed by the *sharpness* of the diffusion target—its perplexity—rather than by the neighborhood size k: a peaked target is captured by a handful of draws no matter how large the graph neighborhood is. The implementation uses weighted sampling without replacement via Gumbel top-k (Efraimidis & Spirakis, 2006; Kool et al., 2019), which draws distinct columns and can only improve coverage. Per anchor and epoch, the student scores $O(m)$ columns instead of the full multi-hop support, and the graph and diffusion targets are computed once, offline.

3.5 The Gauge of Diffusion Matching and Its Reduction

The last question is what the diffusion objective leaves undetermined. Every term in Eq. 1 is a softmax divergence, and a softmax is invariant to a constant shift of its logits.

Proposition 6 (Gauge of the diffusion loss). *$\mathcal{L}_{\text{diff}}$ is invariant under $c^S_{ij} \mapsto c^S_{ij} + a_i$ for arbitrary per-anchor constants a_i. Moreover, if the ambient-scale term of $\mathcal{L}_{\text{diff}}$ vanishes for anchor i on a column set with at least two elements, then $c^S_{ij} = c^T_{ij} + a_i$ on that set for some $a_i \in \mathbb{R}$.*

Distribution matching thus determines all similarity *gaps* but leaves one scalar origin free per supervised row. Downstream uses of embeddings are not invariant to this gauge: a retrieval threshold or a similarity calibration is shared *across* anchors, and per-anchor offsets $\{a_i\}$ reorder cross-anchor comparisons while leaving $\mathcal{L}_{\text{diff}}$ untouched (Appendix A.10).

The freedom is one constant per supervised row, so it shrinks when more rows are supervised and those rows overlap. This is what walk-sampled supervision provides: it constrains rows at non-anchor nodes, and cosine similarity is symmetric, so two rows that each appear in the other's column set cannot carry different offsets.

Definition 1 (Constraint graph). Let $\mathcal{S}$ be the set of supervised rows and, for $u \in \mathcal{S}$, let Ω_u be the column set on which row u is supervised. The constraint graph $\mathcal{G}$ has vertex set $\mathcal{S}$ and an edge $\{u, v\}$ whenever $v \in \Omega_u$ and $u \in \Omega_v$.

Proposition 7 (Gauge reduction by walk supervision). *Assume $\tau_w = \tau_g$ and $|\Omega_u| \geq 2$ for every $u \in \mathcal{S}$, and suppose every supervised term vanishes: the ambient term of $\mathcal{L}_{\text{diff}}$ at each anchor, and the walk term of Eq. 20 at each visited node. Then there are constants $\{a_u\}_{u \in \mathcal{S}}$ with $c^S_{uv} = c^T_{uv} + a_u$ for every $v \in \Omega_u$, and $a_u = a_v$ for every edge $\{u, v\}$ of $\mathcal{G}$. Consequently a is constant on each connected component of $\mathcal{G}$, and the invariance group of the objective is $\mathbb{R}^\kappa$ with κ the number of connected components, rather than $\mathbb{R}^{|\mathcal{S}|}$.*

Two consequences make this the mechanism by which the walk term earns its place. First, the walk builds the edges of $\mathcal{G}$ as it moves: consecutive visited nodes are adjacent in the mutual-kNN graph, mutual adjacency is symmetric, and both endpoints are supervised, so each traversed step contributes an edge. Taking the union of the per-batch constraint graphs over training, $\mathcal{G}$ covers the traversed part of $\mathcal{M}^T_k$, and κ is governed by the connected components of the *teacher graph* rather than by the number of anchors. It is in this sense that the trajectory matters, even though the loss itself does not couple consecutive steps: the path is what chains the constraints.

Second, the residual freedom is benign in a way the per-anchor freedom is not. A single offset shared by an entire component shifts all of its cosines equally, so it preserves every cross-anchor ordering—rank correlations such as those used in semantic textual similarity are exactly invariant to it—and it displaces only the absolute location of a shared decision threshold, a single scalar. Per-anchor offsets have neither property (Remark 6). Section 5.3 tests the resulting prediction: removing the walk term should widen the dispersion of the estimated offsets $\hat{a}_i$ and cost most on tasks that compare cosines across anchors, while leaving within-anchor structure comparatively intact.

4 Methodology

HeatGeo treats the teacher embedding space as a sampled semantic manifold and transfers its geometry through heat diffusion, instantiating the analysis of Section 3. Section 4.1 builds the discrete manifold, Section 4.2 defines a family of diffusion kernels indexed by diffusion time, Section 4.4 trains the student to reproduce all of them at once, and Section 4.5 extends supervision from anchor rows to the rows a random walk visits.

4.1 Teacher Semantic Graph

Let $\mathcal{D} = \{x_i\}_{i=1}^N$ be the unlabeled distillation corpus, and let $t_i = T(x_i)$ denote the cached embedding produced by teacher encoder T. For each example, we retrieve its top-k neighbors under teacher cosine similarity:

$$\mathcal{N}^T_k(i) = \operatorname{TopK}_{j \neq i} \cos(t_i, t_j). \tag{6}$$

To reduce noisy one-directional edges and hubness, HeatGeo retains only mutual kNN connections:

$$j \in \mathcal{M}_k^T(i) \iff j \in \mathcal{N}_k^T(i) \text{ and } i \in \mathcal{N}_k^T(j). \tag{7}$$

The weighted adjacency matrix W^T is

$$W_{ij}^T = \begin{cases} \exp(\cos(t_i, t_j)/\tau_g), & j \in \mathcal{M}_k^T(i), \\ 0, & \text{otherwise,} \end{cases} \tag{8}$$

where τ_g controls graph sharpness. With diagonal degree matrix D^T, the row-normalized transition matrix is

$$P^T = (D^T)^{-1} W^T, \tag{9}$$

and each row P_i^T defines a one-step semantic random-walk distribution from anchor x_i.

4.2 A Multi-Scale Family of Teacher Kernels

Diffusion time is the natural scale parameter of a manifold. On a continuous manifold the heat kernel $h_t(u, v)$ interpolates between the ambient metric as $t \to 0$, where $h_t(u, v) \propto \exp(-d(u, v)^2/4t)$ depends only on the distance between the two points, and global connectivity as t grows, where mass has travelled along the manifold and reflects its intrinsic structure. HeatGeo distills this entire interpolation rather than a single point on it, by pairing the graph walks with the ambient kernel they emerge from.

Graph scales ($r \geq 1$). Powers of the lazy walk $\tilde{P}^T = \frac{1}{2}\big(I + P^T\big)$ expose progressively broader semantic regions,

$$P_r^T = \big(\tilde{P}^T\big)^r, \qquad r \in \{1, 2, 4\}, \tag{10}$$

where the lazy step retains half of the mass in place—the discrete analogue of a heat-semigroup step—and small r preserves sharp local neighborhoods while larger r captures multi-hop connectivity and community structure. Diffusion is computed sparsely by propagating probability mass through graph neighbor lists and retaining the highest-probability columns at each step (Lemma 3).

Ambient scale ($r = 0$). Before any propagation, the teacher metric itself induces a distribution. Writing this small-time limit of the heat kernel in terms of cosine similarity gives

$$p_{i,0}^T(j) = \frac{\exp(\cos(t_i, t_j)/\tau_0)}{\sum_{j'} \exp(\cos(t_i, t_{j'})/\tau_0)}. \tag{11}$$

This scale is defined for every pair in the corpus, whereas the graph scales are supported on the mutual-kNN structure. It is what makes the family an interpolation with two endpoints instead of a set of walks: the graph scales describe how mass *moves* on the manifold, the ambient scale describes the metric it moves on. Because teacher embeddings are already cached for graph construction, Eq. 11 is obtained at no additional cost — no teacher forward pass, no stored hidden states, and no trainable parameters.

We write $\mathcal{R} = \{0, 1, 2, 4\}$ for the full scale family.

4.3 Candidate Sets and Batch-Level Sharing

Distributions over the full corpus are intractable, so each anchor is scored against a candidate set whose induced error Proposition 5 bounds,

$$C_i = C_i^+ \cup C_i^h \cup C_i^r, \tag{12}$$

combining high-probability diffusion neighbors C_i^+, teacher hard negatives C_i^h mined from same-source nearest neighbors outside the graph, and random negatives C_i^r. Candidate sets are redrawn every epoch from precomputed pools, so the student is evaluated against fresh comparisons rather than repeatedly refitting one frozen sample. The graph target at scale $r \geq 1$ is the diffusion mass restricted and renormalized over this set,

$$p_{i,r}^T(j) = \frac{(P_r^T)_{ij}}{\sum_{j' \in C_i} (P_r^T)_{ij'}}, \qquad j \in C_i, \tag{13}$$

which expresses graded semantic relatedness rather than a binary positive–negative label.

Within a mini-batch B, the candidate sets are pooled: every anchor is scored against

$$\mathcal{P}_B = \bigcup_{i \in B} C_i, \tag{14}$$

deduplicated by corpus index and with the anchor's own column removed. Since $\mathcal{P}_B$ is encoded once for the whole batch, this multiplies the comparisons available per anchor by roughly $|B|$ at unchanged encoding cost, and the shared columns couple anchors so that similarity levels become comparable across the batch — the property that semantic textual similarity and a single global cosine threshold both depend on. The ambient scale is normalized over $\mathcal{P}_B$, where Eq. 11 assigns teacher mass to every column, and the graph scales are normalized over C_i, the support on which the walk from anchor i is defined.

4.4 Diffusion Objective

Let $s_i = S_\theta(x_i)$ be the student embedding of anchor x_i. At each scale the student induces a distribution over the corresponding column set $\Omega_{i,r}$ ($\mathcal{P}_B$ for $r = 0$, C_i otherwise),

$$p^S_{i,r}(j) = \frac{\exp(\cos(s_i, s_j)/\tau_r)}{\sum_{j' \in \Omega_{i,r}} \exp(\cos(s_i, s_{j'})/\tau_r)}, \qquad j \in \Omega_{i,r}, \tag{15}$$

and HeatGeo minimizes the weighted divergence to the teacher family,

$$\mathcal{L}_{\text{diff}} = \frac{1}{|B|} \sum_{i \in B} \sum_{r \in \mathcal{R}} \omega_r D_{\text{KL}} \left(p^T_{i,r} \,\|\, p^S_{i,r} \right), \tag{16}$$

with ω_r normalized scale weights, specified below.

The scale measure. The weights discretize a measure over diffusion time. On the dyadic grid $t_j = 2^j$ the spacing is $\Delta t_j = t_j$, so weighting scale r by $\omega_r \propto r^{1-s}$ is the Riemann discretization of dt/t^s. Equal weights per octave, $s = 1$, discretize the scale-invariant measure dt/t — logarithmic time sampling, the convention of heat-kernel signatures (Sun et al., 2009; Bronstein & Kokkinos, 2010; Tsitsulin et al., 2018) and the unweighted dyadic convention of diffusion wavelets (Coifman & Maggioni, 2006) — and we adopt it: the graph scales carry equal weight, so the weights, like the temperatures below, contribute no tuned parameter. The ambient scale is weighted against the graph family as a whole rather than as one more octave: it receives the same total mass as all graph scales combined, splitting the objective evenly between absolute calibration and neighborhood ranking — the two jobs that Sections 4.2 and 4.3 keep on separate column domains.

Resolution-dependent student distributions. Eq. 15 gives each scale its own temperature τ_r, increasing with r so that broader teacher targets are matched at a correspondingly smoother resolution. This is what makes the family informative rather than redundant: by Proposition 3, a shared student distribution reduces the multi-scale objective to a single-scale one against the mixture $\bar{p}^T_i$, so scales that differ only in the teacher would leave the gradient unchanged. Distinct τ_r break this equivalence. Matching the same similarities at several resolutions is over-determined, so the student must reproduce the teacher's cosine *gaps* rather than only its ranking, which is precisely the information that graded similarity tasks measure.

A one-parameter temperature law. Not every member of the ladder is a design choice, and the rest follow one law. The sharpest graph scale's target is exactly the transition row — the lazy walk at $r = 1$ is $\frac{1}{2}(I + P^T)$, whose off-diagonal part renormalizes to P^T_i, itself a softmax of teacher cosines at τ_g — so by Proposition 4 it must be matched at the graph temperature, $\tau_1 = \tau_g$: at any other value the term's zeros are an affine rescaling of the teacher's cosines, incompatible with the ambient scale, and the objective acquires an artificial floor. For the broader scales the heat kernel fixes the functional form: on the unit sphere $d^2 = 2 - 2\cos$, so the small-time kernel $\exp(-d^2/4t)$ of Section 4.2 is a softmax in cosine similarity at temperature $2t$, and diffusion time grows linearly in r, giving $\tau_r \propto r$. We therefore generate the whole ladder from a single exponent,

$$\tau_r = \tau_g \, r^\alpha, \qquad \tau_0 = \tau_g \, r^\alpha_{\max}, \tag{17}$$

where $\alpha = 1$ is the value the heat-kernel argument prescribes — exact when the student embeds the manifold isometrically and the kernel is in its Gaussian regime — and $r = 1$ recovers the tie for every α. The ambient scale sits at the top of the ladder, since it calibrates over the pooled column set, the broadest comparison in the objective, and it uses τ_0 on both the teacher (Eq. 11) and the student (Eq. 15) side, the same-temperature convention of knowledge distillation (Hinton et al., 2015); the walk term is tied to τ_g in Section 4.5. We adopt the heat-kernel value $\alpha = 1$ throughout, so temperatures contribute no tuned parameter at all: the full ladder follows from τ_g and the scale set.

4.5 Walk-Sampled Kernel Matching

Every row of the teacher operator that Eq. 16 matches is indexed by an *anchor*. The rows at the remaining corpus nodes are never supervised, even though a large number of them are already encoded: the pooled set $\mathcal{P}_B$ contains, on every step, several hundred non-anchor texts whose student embeddings are computed and then used only as columns. Walk-sampled kernel matching supervises those rows as well, at no additional encoding cost.

Which rows. From each anchor $i \in B$ we run M random walks of length L on the teacher transition matrix,

$$j_0^{(m)} = i, \qquad j_{t+1}^{(m)} \sim P^T_{j_t^{(m)}}, \qquad t = 0, \dots, L-1, \tag{18}$$

and take the nodes visited at steps $t \geq 1$ as the rows to supervise. Writing ν_B for the normalized visit counts over the batch — the walk's occupancy measure, restricted to nodes present in $\mathcal{P}_B$ — the walk selects rows in proportion to how much diffusion mass reaches them from the batch's anchors, which is the same principle that selects columns in Section 4.3. Visited nodes not already in C_i replace random negatives, so the pooled set covers the walk without enlarging it.

Which columns. For a visited node j the column set is the teacher's own support for that row, intersected with what the batch holds,

$$\Omega_j^w = \mathcal{P}_B \cap \mathcal{M}_k^T(j), \tag{19}$$

and the teacher target is the transition row restricted to it, $P_j^T\big|_{\Omega_j^w}$. Restricting to $\mathcal{M}_k^T(j)$ rather than to all of $\mathcal{P}_B$ is what keeps the term compatible with Section 4.3: on Ω_j^w every column carries teacher mass, so no column receives a target of zero for the accidental reason that it was drawn for a different anchor. A hard negative, in particular, is by construction a high-similarity pair excluded from the mutual-kNN graph, so it can never enter Ω_j^w and is never pushed apart by this term.

The objective. With student distributions $p^S_{w,j}(j') \propto \exp\big(\cos(s_j, s_{j'})/\tau_w\big)$ on Ω_j^w,

$$\mathcal{L}_{\text{walk}} = \sum_j \nu_B(j)\, D_{\text{KL}}\left(P_j^T\big|_{\Omega_j^w} \,\Big\|\, p^S_{w,j}\right), \qquad \tau_w = \tau_g, \tag{20}$$

where tying the student temperature to the graph temperature makes the teacher target exactly attainable: P_j^T is itself a softmax of teacher cosines at τ_g, so Eq. 20 has an infimum of 0 reached precisely when student and teacher cosines agree on Ω_j^w up to a constant — the walk-row instance of Proposition 4, and a tie the implementation enforces rather than exposes. The term is therefore on the same scale as Eq. 16, and λ below trades off two divergences rather than converting units.

Dense targets rather than sampled successors. A natural alternative scores the sampled transitions themselves, $-\log p^S(j_{t+1} \mid j_t)$ summed along each path. Because the walk is first-order and each step is scored on the teacher's own j_t, that path likelihood factorizes into independent one-step terms, and its expectation over the walk is exactly Eq. 20. The sampled form is thus an unbiased but high-variance estimator of the dense one; since P_j^T is already materialized during graph construction, we use the dense form and pay no variance. It also fixes what the term does and does not claim: it matches one-step kernels at walk-visited rows, and nothing in it couples two consecutive steps.

Why step 0 is dropped. The walk starts at the anchor, but the anchor's row is already supervised: the lazy walk at $r = 1$ is $\frac{1}{2}(I + P^T)$, whose off-diagonal part renormalizes to exactly P_i^T. Supervising step 0 would distil the $r = 1$ target a second time, so only $t \geq 1$ contributes.

Full objective.

$$\boxed{\mathcal{L}_{\text{HeatGeo}} = \mathcal{L}_{\text{diff}} + \lambda\, \mathcal{L}_{\text{walk}}} \tag{21}$$

Beyond its role as extra supervision, the walk term removes most of the gauge freedom that Proposition 6 identifies. Each traversed edge constrains two rows that contain each other as columns, and cosine symmetry then forces their similarity offsets to agree; chaining along the walk propagates the identification through the graph, leaving one offset per connected component instead of one per anchor (Proposition 7). This is the sense in which the trajectory, and not merely the set of visited nodes, does work: it is the path that links the constraints.

Training therefore requires only cached teacher embeddings. HeatGeo uses no teacher hidden states, no online teacher forward pass, no task labels, no contrastive or layer-alignment objective, and no change to the student architecture.

5 Experiments

5.1 Experimental Setup

Models. We distill Qwen3-Embedding-4B into BERT-base, a 109M-parameter student encoder. Teacher embeddings are cached once before HeatGeo training. After graph construction, the teacher is removed from GPU memory; student training uses only precomputed diffusion targets and the corresponding candidate texts.

Training Data. Following the TALAS setup (Dao et al., 2026), we train on approximately 15K unlabeled sentences sampled evenly from the three in-domain datasets Emotion, WiC, and STS-B. HeatGeo constructs the teacher semantic graph directly over these texts.

Evaluation Tasks and Metrics. We evaluate on three task families. Text classification uses F1 on Banking77, Tweet, and Emotion. Pair classification uses average precision (AP) on MRPC, SciTail, and WiC. Semantic textual similarity uses Spearman correlation on SICK, STS12, and STS-B. Emotion, WiC, and STS-B are in-domain; the other six datasets are out-of-domain. Avg-In and Avg-Out are the unweighted means over their respective three and six datasets, and Avg. is the unweighted mean over all nine datasets.

Baselines. We compare against the teacher, the undistilled student, SimCSE-unsup, CDM, DSKD, Jasper and Stella, DistillCSE, EMO, and TALAS. The baseline results use the Qwen3-Embedding-4B to BERT-base configuration reported by TALAS (Dao et al., 2026).

Implementation Details. The teacher graph uses $k = 200$ mutual neighbors, and diffusion is taken over the scale family $\mathcal{R} = \{0, 1, 2, 4\}$ with equal weight on the graph scales — the log-uniform scale measure of Section 4.4 — and the ambient scale carrying the same total weight as the graph family combined; like the temperatures, the weights are not tuned. Each anchor draws a candidate set of size 64 from diffusion neighbors, teacher hard negatives, and random negatives, redrawn every epoch, and candidate sets are pooled across mini-batches of 32 anchors. Student temperatures follow the law of Eq. 17 at the heat-kernel value $\alpha = 1$: at $\tau_g = 0.05$ this gives $(\tau_1, \tau_2, \tau_4) = (0.05, 0.10, 0.20)$ and $\tau_0 = 0.20$ on both sides of the ambient scale, with the sharpest scale and the walk term tied to the graph temperature ($\tau_1 = \tau_w = \tau_g$, Proposition 4); no temperature is tuned. Walk-sampled kernel matching draws $M = 4$ walks of length $L = 4$ per anchor per epoch at $\lambda = 0.5$, and is enabled from the second epoch onward so that the student has a neighborhood to rank before its rows are matched. Training runs for five epochs with a learning rate of 3×10^{-5}. Remaining settings are given in the released configuration.

5.2 Main Results

Table 1 reports all three teacher–student settings across the nine datasets. HeatGeo attains the best overall average among compact students in two of the three settings, 77.98 against 76.55 for the strongest baseline when distilling BGE-M3 into MiniLMv2-H768, and 75.33 against 74.79 when distilling Qwen3-Embedding-0.6B into the 22M MiniLMv2-H384. In the Qwen3-Embedding-4B to BERT-base setting it reaches 78.56, ranking second behind TALAS at 78.83.

Table 1: Results across three teacher → student settings on nine datasets. Emotion, WiC, and STS-B (marked with ⋆) are in-domain; the remaining datasets are out-of-domain. Bold and underlined values denote the best and second-best compact-model results within each setting. Baseline results are from TALAS (Dao et al., 2026). For setting (c), we report the epoch checkpoint with the highest overall average.

	Classification (F1)			**Pair Classification (AP)**			**STS (Spearman)**			**Domain Avg**		**Avg.**
	Banking77	Tweet	Emotion⋆	MRPC	SciTail	WiC⋆	SICK	STS12	STS-B⋆	Avg-In	Avg-Out	Avg.
(a) Qwen3-Embedding-4B → BERT-base (109M)												
Teacher	93.63	72.22	68.94	84.79	91.77	66.80	83.43	82.55	86.87	74.20	84.73	81.22
Student base	84.86	47.79	68.02	77.36	67.74	59.22	42.43	21.54	20.30	42.44	60.33	54.36
SimCSE-unsup	89.21	69.13	53.55	82.86	76.59	**71.64**	68.19	61.68	70.36	65.18	74.61	71.47
CDM	89.75	70.58	53.11	84.83	75.19	71.06	69.37	68.48	73.92	66.03	76.37	72.92
DSKD	89.66	69.93	54.51	83.13	77.95	71.53	68.65	63.37	71.42	65.82	75.45	72.24
Jasper and Stella	89.71	73.83	65.98	85.33	80.99	70.50	73.96	67.90	75.65	70.71	78.62	75.98
DistillCSE	90.94	70.90	60.80	82.86	78.98	68.63	75.12	72.61	77.08	68.84	78.57	75.32
EMO	90.78	73.54	67.48	84.46	81.57	69.76	75.66	68.83	77.17	71.47	79.14	76.58
TALAS	91.43	**74.30**	**70.71**	**86.47**	82.66	69.24	**78.38**	**75.40**	80.88	**73.61**	**81.44**	**78.83**
HeatGeo	**91.86**	73.06	69.20	84.72	**86.19**	67.19	77.54	74.19	**83.14**	73.17	81.26	78.56
(b) BGE-M3 → MiniLMv2-H768 (66M)												
Teacher	93.52	73.85	68.56	85.81	91.87	61.47	79.18	78.73	84.87	71.63	83.83	79.76
Student base	81.84	47.70	71.67	79.67	65.06	60.89	49.46	26.76	24.12	44.24	62.41	56.35
SimCSE-unsup	87.89	71.45	55.65	83.87	76.32	67.96	66.93	59.66	63.97	62.53	74.35	70.41
CDM	89.15	70.48	58.71	84.16	75.91	**70.00**	68.14	62.74	67.78	65.50	75.10	71.90
DSKD	89.51	70.53	57.27	84.39	75.36	69.96	68.81	66.78	70.05	65.76	75.90	72.52
Jasper and Stella	88.38	74.13	62.81	85.98	80.33	68.38	74.35	66.80	74.55	68.58	78.33	75.08
DistillCSE	90.50	72.70	59.88	84.39	78.48	69.49	73.97	70.00	73.73	67.70	78.34	74.79
EMO	90.19	74.26	61.48	85.31	80.82	68.30	75.41	67.06	76.35	68.71	78.84	75.46
TALAS	90.62	**74.81**	63.30	**87.05**	81.94	66.28	76.86	69.04	79.01	69.53	80.05	76.55
HeatGeo	**90.86**	73.08	**69.50**	85.74	**83.57**	65.77	**77.32**	**74.18**	**81.76**	**72.34**	**80.79**	**77.98**
(c) Qwen3-Embedding-0.6B → MiniLMv2-H384 (22M)												
Teacher	93.21	71.33	66.53	83.37	90.00	66.85	80.64	77.11	84.60	72.66	82.61	79.29
Student base	68.29	41.07	69.90	78.39	64.56	59.58	47.60	21.95	22.08	40.91	58.45	52.60
SimCSE-unsup	86.67	69.56	54.00	82.29	73.76	65.71	66.04	59.11	62.08	60.60	72.91	68.80
CDM	86.07	70.32	53.04	81.97	72.97	68.34	65.97	61.12	65.75	62.38	73.07	69.51
DSKD	85.66	69.86	52.25	82.17	73.52	68.58	66.54	62.52	66.70	62.51	73.38	69.76
Jasper and Stella	85.94	71.25	57.99	83.36	76.17	68.25	70.57	63.81	70.11	65.45	75.18	71.94
DistillCSE	88.03	69.87	54.24	83.20	77.66	67.88	70.09	68.19	71.53	64.55	76.17	72.30
EMO	87.03	71.67	59.63	83.45	78.17	**68.89**	70.81	65.78	71.42	66.65	76.15	72.98
TALAS	86.69	**72.77**	60.94	**85.10**	**81.42**	66.65	72.51	70.77	76.22	67.94	**78.21**	74.79
HeatGeo	**88.31**	70.41	**67.13**	83.00	80.60	65.51	**73.98**	**72.14**	**76.90**	**69.85**	78.07	**75.33**

The advantage is concentrated where the objective operates. HeatGeo gives the best student result on Banking77 and on STS-B in all three settings, improving STS-B by 2.26, 2.75, and 0.68 Spearman points over the strongest baseline, and it leads the STS family on all three datasets in both MiniLMv2 settings. It also obtains the best in-domain average in the two settings it wins overall, by 2.81 and 1.91 points, with the largest single-dataset gains on Emotion (+6.20 and +6.19 F1). The margins grow as the student shrinks: matching diffusion distributions rather than individual teacher coordinates asks the student to preserve relative neighborhood structure, which remains attainable when embedding capacity is well below the teacher's. WiC is the one dataset where HeatGeo trails consistently, reflecting that word-in-context judgments depend on token-level sense disambiguation rather than the sentence-level neighborhood geometry the diffusion targets describe.

5.3 Theory-Driven Analysis

Each analysis targets a specific prediction of Section 3. Unless stated otherwise, all runs use the BGE-M3 → MiniLMv2-H768 setting and report the same nine-dataset protocol as Table 1.

Graph size k. Theorem 1 transfers geometry from whichever graph the teacher induces, and the manifold interpretation behind it improves as neighborhoods grow—until neighborhoods become so large that they cut across semantic boundaries and add noise edges. The theory therefore predicts a rise followed by a plateau or mild decline as k grows, rather than monotone improvement. Table 2 sweeps the graph size.

Table 2: Effect of graph size k (BGE-M3 $\rightarrow$ MiniLMv2-H768). Default in bold column header.

	$k = 50$	$k = 100$	$\mathbf{k = 200}$	$k = 400$
Avg-In	**[TBD]**	**[TBD]**	72.34	**[TBD]**
Avg-Out	**[TBD]**	**[TBD]**	80.79	**[TBD]**
Avg.	**[TBD]**	**[TBD]**	77.98	**[TBD]**

Candidate size m. Proposition 5 bounds the error of scoring each anchor on a sampled candidate set by the uncovered target mass, which decays with m at a rate set by the sharpness of the diffusion target rather than by k. Because the diffusion targets are far more peaked than uniform over their support, performance should saturate at a candidate size well below the neighborhood size, and enlarging m past that point should mainly add compute. Table 3 sweeps the candidate size.

Table 3: Effect of candidate size m (BGE-M3 $\rightarrow$ MiniLMv2-H768). Default in bold column header.

	$m = 16$	$m = 32$	$\mathbf{m = 64}$	$m = 128$	$m = 256$
Avg-In	**[TBD]**	**[TBD]**	72.34	**[TBD]**	**[TBD]**
Avg-Out	**[TBD]**	**[TBD]**	80.79	**[TBD]**	**[TBD]**
Avg.	**[TBD]**	**[TBD]**	77.98	**[TBD]**	**[TBD]**

Walk supervision and the similarity gauge. Propositions 6 and 7 make a sharp, falsifiable prediction. Without $\mathcal{L}_{\text{walk}}$ the supervised rows are the anchors alone, the constraint graph has no edges, and one similarity offset per anchor is free; with it, each traversed edge identifies two offsets and the freedom collapses to one constant per connected component of the teacher graph. Tasks that compare cosines *across* anchors against a shared scale—pair classification with a global threshold, and cross-anchor rank correlation in semantic textual similarity—should therefore degrade when the term is removed, while within-anchor structure such as classification by nearest neighborhoods should be comparatively unaffected. Table 4 removes the term and additionally reports the dispersion of the estimated per-anchor offsets $\hat{a}_i = \sum_j p^T_{i,0}(j)\big[\cos(s_i, s_j) - \cos(t_i, t_j)\big]$, which the theory predicts to widen without walk supervision, together with κ, the number of components of the realized constraint graph.

Table 4: Effect of removing walk-sampled kernel matching (BGE-M3 $\rightarrow$ MiniLMv2-H768). Offset dispersion is the standard deviation of $\hat{a}_i$ across anchors; κ is the number of connected components of the realized constraint graph.

	Class. (F1)	Pair (AP)	STS (Spear.)	Avg.	Offset disp.	κ
HeatGeo	**[TBD]**	**[TBD]**	**[TBD]**	77.98	**[TBD]**	**[TBD]**
w/o $\mathcal{L}_{\text{walk}}$	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**

How much walk supervision. The walk controls two quantities that the theory separates: how many rows are supervised, set by M and L, and how strongly they are weighted against the anchor rows, set by λ. Because Eq. 20 is a divergence with infimum 0 on the same scale as Eq. 16, λ is a genuine trade-off, and the prediction is a plateau rather than a monotone trend: once the traversed edges connect the graph, further walks add rows but no new identification. Table 5 sweeps both.

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Table 5: Walk length L and weight λ (BGE-M3 $\rightarrow$ MiniLMv2-H768), reporting the nine-dataset average. Defaults in bold column headers.

	Walk length L			
	$L = 2$	$\mathbf{L = 4}$	$L = 6$	$L = 8$
Avg.	**[TBD]**	77.98	**[TBD]**	**[TBD]**
	Walk weight λ			
	$\lambda = 0.1$	$\lambda = 0.3$	$\lambda = \mathbf{0.5}$	$\lambda = 1.0$
Avg.	**[TBD]**	**[TBD]**	77.98	**[TBD]**

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A Proofs

Throughout, $\|v\|_1 = \sum_j |v_j|$ and $\mathrm{TV}(p,q) = \frac{1}{2}\|p-q\|_1$. For a nonnegative vector v with $\|v\|_1 > 0$ and a subset C of its support, $v|_C$ denotes restriction to C followed by renormalization.

A.1 Proof of Proposition 1 (neighborhood coverage)

Within an epoch the corpus is partitioned uniformly at random into batches of size B. By exchangeability, the batch containing i contains $B-1$ of the remaining $N-1$ examples, uniformly without replacement, so for any fixed $j \neq i$,

$$\Pr[j \in \mathcal{B}] = \frac{B-1}{N-1}. \tag{22}$$

(i) follows by linearity over the k neighbors. (ii) is Markov's inequality applied to the nonnegative count $|\mathcal{B} \cap \mathcal{N}_k(i)|$: $\Pr[\text{count} \geq 1] \leq \mathbb{E}[\text{count}]$. For (iii), epochs are independent, so with $p_e = (B-1)/(N-1)$ the probability that a given neighbor pair co-occurs within T epochs is $1-(1-p_e)^T \leq Tp_e$, and the expected fraction of the k neighbor pairs of anchor i observed within T epochs is the same bound by linearity. □

A.2 Proof of Proposition 2 (bias of in-batch targets) and the margin condition

Condition on the event $E_0 = \{\mathcal{B} \cap \mathcal{N}_k(i) = \emptyset\}$. On E_0 the in-batch target $p_i^{T,\mathcal{B}}$ is supported on $\mathcal{B} \setminus \{i\} \subseteq \mathcal{D} \setminus \mathcal{N}_k(i)$, hence assigns mass 0 to $\mathcal{N}_k(i)$, while $p_{i,0}(\mathcal{N}_k(i)) \geq 1-\eta$ by Assumption 1. Since $\mathrm{TV}(p,q) \geq |p(A) - q(A)|$ for any event A, taking $A = \mathcal{N}_k(i)$ gives $\mathrm{TV}(p_i^{T,\mathcal{B}}, p_{i,0}) \geq 1-\eta$ on E_0. Therefore

$$\mathbb{E}_{\mathcal{B}}\big[\mathrm{TV}\big(p_i^{T,\mathcal{B}}, p_{i,0}\big)\big] \geq (1-\eta)\Pr[E_0] \geq (1-\eta)\Big(1 - \tfrac{(B-1)k}{N-1}\Big), \tag{23}$$

using Proposition 1(ii). □

Margin condition for Assumption 1. Suppose $\min_{j \in \mathcal{N}_k(i)} c_{ij}^T \geq \max_{j \notin \mathcal{N}_k(i) \cup \{i\}} c_{ij}^T + \Delta$. Each non-neighbor's unnormalized weight is at most $e^{-\Delta/\tau_0}$ times each neighbor's, so

$$\frac{p_{i,0}(\mathcal{D} \setminus \mathcal{N}_k(i))}{p_{i,0}(\mathcal{N}_k(i))} \leq \frac{(N-1-k)\,e^{-\Delta/\tau_0}}{k}, \tag{24}$$

and $\eta \leq (N/k)e^{-\Delta/\tau_0}$, which is at most a prescribed η whenever $\tau_0 \leq \Delta/\log\big(N/(k\eta)\big)$.

A.3 Proof of Lemma 1 (one step to r steps)

Write $\tilde{P} - \tilde{Q} = \frac{1}{2}(P - Q)$ and telescope:

$$\tilde{P}^r - \tilde{Q}^r = \sum_{s=0}^{r-1} \tilde{P}^s \,(\tilde{P} - \tilde{Q})\, \tilde{Q}^{\,r-1-s}. \tag{25}$$

Fix a row i and a summand s. The vector $\mu = \delta_i^\top \tilde{P}^s$ is a probability vector, so $\mu(\tilde{P} - \tilde{Q})$ is a convex combination of the rows of $\tilde{P} - \tilde{Q}$ and

$$\big\|\mu(\tilde{P} - \tilde{Q})\big\|_1 \le \max_j \big\|\tfrac{1}{2}(P_j - Q_j)\big\|_1 = \max_j \mathrm{TV}(P_j, Q_j) \le \varepsilon. \tag{26}$$

Right-multiplication by a row-stochastic matrix does not increase the ℓ_1 norm: $\|\nu \tilde{Q}^{\,r-1-s}\|_1 \le \|\nu\|_1$. Summing the r terms gives $\|\delta_i^\top(\tilde{P}^r - \tilde{Q}^r)\|_1 \le r\varepsilon$, i.e. $\mathrm{TV} \le r\varepsilon/2$. □

A.4 Proof of Lemma 2 (restriction)

Split the ℓ_1 distance over C and $\bar{C}$. Outside C, $\sum_{\bar{C}} |p_j - q_j| \le p(\bar{C}) + q(\bar{C}) \le \delta_T + \delta_S$. Inside C, write $p_j = p(C)\, p|_C(j)$ and $q_j = q(C)\, q|_C(j)$:

$$\sum_C \big|p_j - q_j\big| \le p(C) \sum_C \big|p|_C(j) - q|_C(j)\big| + \big|p(C) - q(C)\big| \le 2\,\mathrm{TV}\big(p|_C, q|_C\big) + \delta_T + \delta_S. \tag{27}$$

Adding the two parts and halving yields the claim. □

A.5 Leakage Control (Remark 1)

We first verify the inequality stated in Remark 1. Let $\tilde{p} = p|_C$ and $\tilde{q} = q|_C$ denote the restricted teacher and student distributions on the candidate set $C = C^+ \cup C^-$, and suppose $D_{\mathrm{KL}}(\tilde{p} \,\|\, \tilde{q}) \le \varepsilon$ with $\tilde{p}(C^-) \le \varepsilon_{\mathrm{neg}}$. The partition $C = C^+ \cup C^-$ induces binary distributions $(\tilde{p}(C^+), \tilde{p}(C^-))$ and $(\tilde{q}(C^+), \tilde{q}(C^-))$, and by the data-processing inequality

$$D_{\mathrm{KL}}\big(\tilde{p} \,\|\, \tilde{q}\big) \ge d_{\mathrm{KL}}\big(\tilde{p}(C^+) \,\|\, \tilde{q}(C^+)\big), \tag{28}$$

where d_{KL} is the binary divergence. Pinsker's inequality for the binary pair gives $|\tilde{p}(C^+) - \tilde{q}(C^+)| \le \sqrt{\varepsilon/2}$, hence $\tilde{q}(C^-) \le \tilde{p}(C^-) + \sqrt{\varepsilon/2} \le \varepsilon_{\mathrm{neg}} + \sqrt{\varepsilon/2}$. □

Remark 2 (From sampled negatives to the full complement). The bound above controls the student's mass on the negative columns actually drawn. The random negatives are uniform draws from the complement of the diffusion support, redrawn every epoch, so over training the student's scores are constrained on a fresh uniform sample of the complement at every epoch: by exchangeability, a student that hides a constant fraction of softmax mass on some subset of the complement would place mass on the drawn negatives in expectation proportional to that fraction, contradicting a uniformly small restricted loss. A quantitative uniform-convergence version of this argument (a bound on δ_S in expectation over the negative-sampling process) is deferred to the final version.

A.6 Proof of Theorem 1 (local-to-global transfer)

Fix an anchor i. Pinsker's inequality applied to the restricted divergence gives $\mathrm{TV}(p_{i,1}|_{C_i}, q_{i,1}) \le \sqrt{\varepsilon/2}$. Lemma 2 with $p = p_{i,1}$, q the student's full row, and $C = C_i$ then bounds the full one-step rows:

$$\mathrm{TV}\big(P_i^T,\, P_i^S\big) \le \sqrt{\varepsilon/2} + \delta_T + \delta_S =: \varepsilon_{\mathrm{row}}. \tag{29}$$

Applying Lemma 1 with row-wise bound $\varepsilon_{\mathrm{row}}$ yields, for every $r \ge 1$, $\max_i \mathrm{TV}\big(\delta_i^\top(\tilde{P}^T)^r, \delta_i^\top(\tilde{P}^S)^r\big) \le r\,\varepsilon_{\mathrm{row}}/2$. □

Corollary 1 is the same argument applied directly to the scale-r family: Pinsker plus Lemma 2 bound the r-step rows without invoking Lemma 1, so no factor of r appears.

Remark 3 (Average-case variant). Training controls the divergence averaged over anchors rather than its maximum. By Jensen's inequality, $\frac{1}{N}\sum_i \mathrm{TV}(p_{i,1}|_{C_i}, q_{i,1}) \le \sqrt{\bar{\varepsilon}/2}$ with $\bar{\varepsilon}$ the average restricted divergence, and the telescoping argument of Lemma 1 can be carried out under any fixed initial distribution over anchors, giving the same bounds for anchor-averaged errors.

A.7 Proof of Proposition 3 (collapse)

With one shared student distribution p_i^S,

$$\sum_r \omega_r D_{\mathrm{KL}}\big(p_{i,r}^T \,\|\, p_i^S\big) = \sum_r \omega_r \Big[H\big(p_{i,r}^T, p_i^S\big) - H\big(p_{i,r}^T\big)\Big] = H\Big(\textstyle\sum_r \omega_r p_{i,r}^T,\, p_i^S\Big) - \textstyle\sum_r \omega_r H\big(p_{i,r}^T\big), \tag{30}$$

using that cross-entropy $H(p,q) = -\sum_j p_j \log q_j$ is linear in its first argument and $\sum_r \omega_r = 1$. The subtracted entropy term does not depend on the student. □

A.8 Proof of Proposition 4 (temperature ties)

(i) If the divergence vanishes, the two softmax distributions coincide on Ω, and taking log-ratios of any two columns $j, j' \in \Omega$ (as in Appendix A.10) gives

$$\frac{c_{ij}^S - c_{ij'}^S}{\tau_S} = \frac{c_{ij}^T - c_{ij'}^T}{\tau}, \tag{31}$$

so $c_{ij}^S - (\tau_S/\tau)\, c_{ij}^T$ is constant on Ω; call it a_i. Conversely, any score assignment of that form produces identical softmax distributions and zero divergence, so the infimum 0 is attained exactly on the stated family.

(ii) The ambient term carries the same temperature τ_0 on both sides, so its vanishing gives, by the shift case of (i) (equivalently Proposition 6), $c_{ij}^S = c_{ij}^T + b_i$ on Ω_0. Combining with (i) on the overlap,

$$\left(1 - \frac{\tau_S}{\tau}\right) c_{ij}^T = a_i - b_i \qquad \text{for every } j \in \Omega \cap \Omega_0, \tag{32}$$

and evaluating at two columns with $c_{ij}^T \neq c_{ij'}^T$ forces $\tau_S = \tau$. □

A.9 Truncated Walks and Proof of Proposition 5

The diffusion targets are computed by a truncated walk; its error is additive across steps.

Lemma 3 (Truncation error)**.** *If the walk is computed with per-step truncation to the highest-mass columns followed by renormalization, dropping mass fraction δ_s at step s, then after r steps the computed distribution $\hat{p}$ satisfies $\mathrm{TV}(\hat{p}, p) \leq \sum_{s \leq r} \delta_s$.*

Proof of Lemma 3. First, for any nonnegative vector v with total mass 1 and truncation $R(v)$ that zeroes a set of mass δ and renormalizes,

$$\mathrm{TV}\big(R(v), v\big) = \tfrac{1}{2}\Big[\sum_{\text{kept}} \Big(\tfrac{v_j}{1-\delta} - v_j\Big) + \sum_{\text{dropped}} v_j\Big] = \tfrac{1}{2}\big[\delta + \delta\big] = \delta. \tag{33}$$

Let v_s denote the exact walk after s steps and $\hat{v}_s$ the truncated walk, with $M = \tilde{P}$ the (row-stochastic) step operator and δ_s the mass fraction dropped at step s. Then

$$\mathrm{TV}(\hat{v}_s, v_s) \leq \mathrm{TV}\big(R(\hat{v}_{s-1}M), \hat{v}_{s-1}M\big) + \mathrm{TV}\big(\hat{v}_{s-1}M, v_{s-1}M\big) \leq \delta_s + \mathrm{TV}(\hat{v}_{s-1}, v_{s-1}), \tag{34}$$

using the triangle inequality and that a stochastic matrix does not increase total variation. Induction over $s \leq r$ gives $\mathrm{TV}(\hat{v}_r, v_r) \leq \sum_{s \leq r} \delta_s$. □

Proof of Proposition 5. The identity $\mathrm{TV}(p|_C, p) = p(\bar{C})$ follows from the first display above with $\delta = p(\bar{C})$. For m independent draws from p,

$$\mathbb{E}\big[p(\bar{C})\big] = \sum_j p_j\, (1 - p_j)^m. \tag{35}$$

Fix $t > 0$ and split the sum: atoms with $p_j \geq t$ contribute at most $(1-t)^m \sum_j p_j \leq e^{-tm}$, and atoms with $p_j < t$ contribute at most $\mathrm{Tail}_p(t)$. Minimizing over t gives the bound. If every atom of the effective support carries mass at least c/ρ, take $t = c/\rho$: the tail term vanishes and the bound is $e^{-cm/\rho}$. □

Remark 4 (Sampling without replacement). The implementation draws candidates by Gumbel top-k, which realizes weighted sampling without replacement (Efraimidis & Spirakis, 2006; Kool et al., 2019). Draws are distinct, so after m draws the covered mass is at least that of m independent draws in the natural coupling that re-draws duplicates; the bound of Proposition 5 therefore remains valid.

A.10 Proofs of Propositions 6 and 7 (gauge and its reduction)

Invariance. For any temperature τ and column set Ω, $\mathrm{softmax}\big((c_i + a_i)/\tau\big)_{j\in\Omega} = \mathrm{softmax}\big(c_i/\tau\big)_{j\in\Omega}$, since the constant $e^{a_i/\tau}$ cancels between numerator and normalizer. Every term of $\mathcal{L}_{\text{diff}}$ is a divergence between a fixed teacher target and such a softmax, so the loss is unchanged.

Zero set. Suppose the ambient-scale term vanishes for anchor i on the column set Ω, $|\Omega| \geq 2$: then $\mathrm{softmax}(c_i^S/\tau_0) = \mathrm{softmax}(c_i^T/\tau_0)$ on Ω. Taking log-ratios of any two columns $j, j' \in \Omega$,

$$\frac{c_{ij}^S - c_{ij'}^S}{\tau_0} = \log\frac{p_{i,0}(j)}{p_{i,0}(j')} = \frac{c_{ij}^T - c_{ij'}^T}{\tau_0}, \tag{36}$$

so $c_{ij}^S - c_{ij}^T$ is constant in j on Ω; call it a_i. This proves Proposition 6.

Walk rows carry the same form of constraint. The teacher's transition row is itself a softmax of teacher cosines at the graph temperature, $P_j^T(j') \propto \exp(c_{jj'}^T/\tau_g)$ on $\mathcal{M}_k^T(j)$, and restricting a softmax to a subset and renormalizing is the softmax over that subset. Hence $P_j^T|_{\Omega_j^w} = \mathrm{softmax}_{\Omega_j^w}(c_{j\cdot}^T/\tau_g)$, and the walk term of Eq. 20 vanishes at j exactly when $\mathrm{softmax}_{\Omega_j^w}(c_{j\cdot}^S/\tau_w) = \mathrm{softmax}_{\Omega_j^w}(c_{j\cdot}^T/\tau_g)$. With $\tau_w = \tau_g$ the log-ratio argument above applies verbatim on Ω_j^w, giving $c_{jv}^S = c_{jv}^T + a_j$ for all $v \in \Omega_j^w$.

Symmetry closes the gauge along edges. Every supervised row $u \in \mathcal{S}$ therefore satisfies $c_{uv}^S = c_{uv}^T + a_u$ on Ω_u, whether it was supervised by the ambient term or by the walk term. Let $\{u, v\}$ be an edge of the constraint graph of Definition 1, so $v \in \Omega_u$ and $u \in \Omega_v$. Applying the identity in both directions,

$$c_{uv}^S = c_{uv}^T + a_u, \qquad c_{vu}^S = c_{vu}^T + a_v. \tag{37}$$

Cosine similarity is symmetric, $c_{uv}^S = c_{vu}^S$ and $c_{uv}^T = c_{vu}^T$, so subtracting gives $a_u = a_v$. A function constant across every edge is constant on every connected component, so a takes at most κ distinct values, one per component of $\mathcal{G}$. Conversely, adding a common constant to all cosines within a component leaves every softmax in the objective unchanged, so the invariance group is exactly $\mathbb{R}^\kappa$. This proves Proposition 7. □

Remark 5 (The walk lays down the edges). Consecutive nodes j_t, j_{t+1} of a walk with $t \geq 1$ are adjacent in $\mathcal{M}_k^T$, which is symmetric by construction, and both are supervised rows whenever both lie in $\mathcal{P}_B$. Then $j_{t+1} \in \Omega_{j_t}^w$ and $j_t \in \Omega_{j_{t+1}}^w$, so the step is an edge of $\mathcal{G}$: a walk of length L contributes a path of $L-1$ edges. Without the walk term $\mathcal{S}$ is the set of anchors and $\mathcal{G}$ has no edges unless two anchors happen to appear in each other's pooled columns, so $\kappa = |\mathcal{S}|$ and the gauge is one constant per anchor.

Remark 6 (Downstream tasks are not gauge-invariant). Consider two anchors i, i' and a downstream decision that thresholds cosine similarity at a shared value θ, as in pair classification or a global retrieval cutoff. Offsets $a_i = +\Delta$, $a_{i'} = -\Delta$ leave $\mathcal{L}_{\text{diff}}$ unchanged by Proposition 6, yet move pairs of i above the threshold and pairs of i' below it, changing the decisions. Rank correlations computed across anchors, as in semantic textual similarity, are affected in the same way. The gauge is therefore task-relevant. A single offset shared across a component is the benign case: it shifts every cosine in the component equally, so it preserves all orderings and moves only the absolute location of θ.